PREP 4, LESSONS 7–8

Your investigation

Planning a fair test; dealing with odd repeats; evaluating and improving a method

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Explain why control variables must be kept the same. [1]

2 State how uncertainty is calculated from a set of repeated readings. [1]

3 What is an anomalous result? [1]

Key ideas LESSONS 7–8

Show your working

4 A group investigates how the **length of a pendulum** affects the **time for 10 swings**.

a Name two control variables. [1]

b Explain why they time 10 swings instead of one. [1]

5 One row of their results:

Length / cm	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
40	12.6	12.8	15.9	

a Which trial is anomalous? Trial 1 | Trial 2 | Trial 3 [1]

b Calculate the mean, leaving out the anomalous result. Write it in the table. [1]

6 Evaluate the method: give **one** improvement, and explain how it makes the results better. [2]

7 Their graph is a curve. Should they join the points dot-to-dot? yes | no Give the reason. [1]

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